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Astro Project

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ABSTRACT

Abstract

1. INTRODUCTION

2. LITERATURE REVIEW

3. METHODOLOGY

4. RESULTS

The study evaluates the potential of Gaia DR3 epoch photometry and TESS light curves to identify eclipsing binaries that are high-priority targets for the PLATO mission can observe. A selection of eclipsing binary systems has been analysed, looking at their physical parameters, characteristics and alignment with PLATO's detection capabilities.

The sample was chosen using a minimum orbital period threshold of 2.5 days, minimising any ellipsoidal variables in the light curves. Although most binary systems had a slight ellipsoidal distortion, they were well-suited for analysis. The selected systems have orbital periods ranging from 2.76 to 17.55 days, with primary masses being between 0.9 and 2 solar masses and secondary masses between 0.6 and 1.9 solar masses, and the Temperatures ranging from 6100 to 11000 Kelvins. These properties make them suitable for studying their stellar evolution and binary interactions.

Shorter orbital periods exhibit evidence of tidal interaction, leading to a circular orbit confirming theoretical predictions. Justesen & Albrecht (2021) highlight how stellar temperature and orbital distance influence tidal circularisations. The orbital period range obtained shows star systems within 3-8 days have a secondary eclipses occur at phase approximately 0.5, confirming they are near circular orbit. while the star systems below or above that threshold reveal eccentric behaviour.

Star systems outside this threshold proved to have eccentric characteristics. TIC 201497357 with an orbital period below 3 days, and TIC 7695666 and TIC 78568736 with periods exceeding 8 days required eccentricity parameters to be adjusted to achieve a good fit, aligning with Justesen & Albrecht (2021) that binaries with very short or large separations have higher levels of eccentricity.

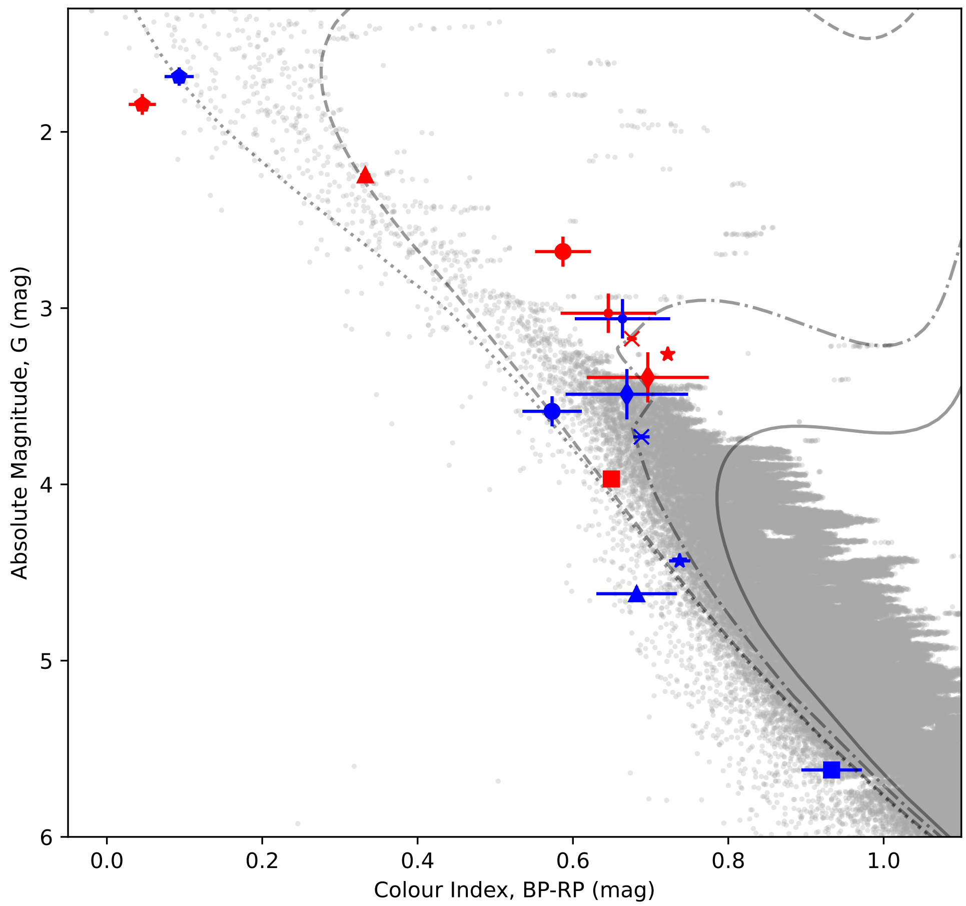

Figure 1. Enter Caption

5. CONCLUSION

Facilities:

Software: `jktebop| v43` (Southworth 2013), `Lightcurve| v2.5.0` (Lightkurve Collaboration et al. 2018),

APPENDIX

APPENDIX

REFERENCES

Justesen, A. B., & Albrecht, S. 2021, The Astrophysical Journal, 912, 123. https://dx.doi.org/10.3847/1538-4357/abefcd

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Southworth, J. 2013, A&A, 557, A119, doi: 10.1051/0004-6361/201322195